Ashmit Mukherjee

New York University Abu Dhabi

asm8879@nyu.edu | ashmitmukherjee.com | LinkedIn | GitHub

Education

New York University Abu Dhabi *Expected May 2027*

B.S. in Computer Science. Studied at NYU Abu Dhabi, NYU Paris, and NYU New York.

Relevant coursework: Natural Language Processing; Principles of Data Science; Statistics for Social & Behavioral Sciences; Data Structures & Algorithms; Computer Systems Architecture; Operating Systems.

Research Interests

I build and evaluate AI systems using data, statistical inference, machine learning, experimental design, and research engineering. My work spans human-AI collaboration, parameter-efficient model adaptation, and multilingual NLP.

Publications and Working Papers

Benjamin Rosche, **Ashmit Mukherjee**, and Hanan Salam. "AI and Human Collaboration in Social Dilemmas: Strategic Interpretation, Normative Interpretation, and Collective Action in Shared Supply Chains." *Working paper*, June 2026.

Ashmit Mukherjee and collaborators. "Bio-Informed LoRA for Signal Peptide Prediction." *Manuscript in preparation*, 2026.

Research Experience

Research Assistant: Human-AI Collaboration, NYU Abu Dhabi *Feb 2026–Present*

- Co-developing an online experiment on how AI-mediated interpretation and coordination affect collective action in shared global supply chains, with Benjamin Rosche and Hanan Salam.
- Designed the study's repeated decision setting, AI-support conditions, and behavioral measurement plan.
- Specified controls and outcomes for evaluating cooperation, responsibility attribution, and coordination over time.

Research Assistant: eBRAIN Lab, NYU Abu Dhabi *Jan 2026–Present*

- Implemented and evaluated **Bio-Informed LoRA**, a parameter-efficient adaptation method that injects BLOSUM62, hydrophobicity, and Grantham priors into LoRA adapters for ESM-2 signal peptide prediction.
- Found BLOSUM62-guided LoRA to be the strongest mean performer among evaluated methods at 650M; model-scale reruns made clear that full fine-tuning is favored at 150M and on 3B cleavage metrics.
- Designed multi-seed evaluation protocols and a one-factor sensitivity screen showing that several single-seed improvements did not replicate.
- Manuscript in preparation (see Writing).

Research Projects

Hinglish Named Entity Recognition Benchmark *Fall 2024*

- Fine-tuned multilingual transformer models, mBERT and XLM-RoBERTa, on the COMI-LINGUA dataset for Hindi-English code-mixed NER.
- Compared against zero-shot GPT-4o and Claude-3.5-Sonnet baselines under a common evaluation protocol; fine-tuned XLM-RoBERTa achieved 78% entity-level F1 versus 76% for GPT-4o.
- Analyzed trade-offs between model scale, task-specific fine-tuning, generalization, and computational cost.

Data Science Salary Prediction Platform *Fall 2025*

- Built a benchmarking pipeline for more than 15 regression models using over 3,000 job postings; tracked experiments with MLflow and compared MAE, RMSE, and R^2.
- Used SHAP to interpret associations between model predictions and features including geography, firm size, and skill requirements.

Community-Level Crime Modeling *Spring 2025*

- Modeled statistical associations between socioeconomic indicators and violent crime rates across 1,994 communities; the analysis does not make causal claims.
- Cross-validation yielded $R^2 = 0.85$ for the selected predictive model; presented results through an interactive Streamlit interface.

CAMP: Campus Asset Management Platform *Fall 2025*

- Contributed to a team-built platform for campus resource allocation and scheduling under shared constraints, including role-based access and conflict resolution.
- Built RESTful APIs, integration testing, containerized deployment, and automated CI/CD workflows.

Industry Experience

Machine Learning Engineer Intern, Zeek *Jun–Aug 2025*

- Developed and evaluated segmentation models on noisy real-world data, iterating on preprocessing and model selection for production reliability.
- Applied model interpretability techniques to study robustness and bias sensitivity, helping the team decide which model variants to deploy.
- Contributed to the team's evaluation workflow by adding standardized metrics and validation procedures to the model development cycle.

Technical Skills

Languages: Python, C++, C#, JavaScript
ML & NLP: PyTorch, Transformers (Hugging Face), Scikit-Learn, XGBoost, PyCaret, SHAP
Computational Modeling: NumPy, NetworkX, game-theoretic simulation
Data Analysis: Pandas, SciPy, Matplotlib, Seaborn, Plotly
Experimental Methods: Experimental design, regression, cross-validation, multi-seed evaluation, sensitivity analysis
Research Engineering: MLflow, Git, Docker, GitHub Actions, REST APIs, automated testing, MongoDB
Applications: Streamlit, React, Node.js, LaTeX